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External validation of AI assisted colposcopy using WHO dataset for cervical precancer and cancer detection

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Artificial intelligence (AI)-guided colposcopy has previously been developed to assist colposcopists in detecting cervical intraepithelial neoplasia grade 2 or worse (CIN2 +) and guiding biopsies, but internal testing of the AI model could overestimate diagnostic performance. We validated the performance of the AI-guided colposcopy on an independent WHO open-source set of 187 patients with 855 colposcopic images. Forty-five colposcopists with different levels of experience from 12 regions in China participated in the validation set. The sensitivity of the AI system on CIN2+ standalone diagnosis were 84.2% (95%CI 74.4–90.7%) in 187 cases. With AI assistance, colposcopist sensitivity increased from 84.8% (95% CI, 82.7–86.9%) to 90.6% (95% CI, 88.7–92.1%). The improvement was most evident among low-experienced colposcopists, with sensitivity increasing by 6.5%, and AUC from 72.4% (95% CI, 67.5–77.3%) to 76.3% (95% CI, 72.3–80.3%; $p = 0.043$). AI-guided biopsy support was also associated with a reduction in the mean number of biopsies per case, from 2.48 to 2.02. These findings highlight the AI system's robust performance on an independent external dataset and its capacity to bridge the diagnostic experience gap among readers, supporting its clinical implementation as a valuable adjunct for colposcopic assessment.

Colposcopy plays a fundamental role in the cervical cancer screening for triage of women with abnormal cytology and/or high-risk human papillomavirus (hr-HPV) infections, significantly contributing to the enhancement of early diagnostic and treatment rates. However, it is highly influenced by subjective visual interpretation and guided biopsies[1]. Accurate colposcopy is heavily dependent upon the operator's expertise. Colposcopy services remain significant challenges due to low examination reproducibility and lack of standardization in colposcopy equipment and terminology[2,3]. The inaccurate colposcopy examination may result in increased misdiagnosis rates and delayed treatment initiation for patients with high-grade squamous intraepithelial lesions (HSIL), ultimately increasing the risk of cancer progression[4]. Furthermore, the diagnostic uncertainty stemming from inadequate colposcopic skills among colposcopists in various clinics often leads to additional unnecessary biopsies and diagnostic procedures, exacerbating the strain on already limited healthcare resources.

As a solution for subjective medical imaging interpretation, artificial intelligence (AI) is often designed as a clinical decision support tool due to its comparable diagnostic accuracy. In recent years, multiple AI-guided colposcopy systems for cervical precancer and cancer detection have been developed[5–9]. A standalone AI system has demonstrated sensitivity for detecting high-grade lesions comparable to colposcopy[10], highlighting its potential to address challenges in resource-limited healthcare settings. However, most of these studies rely on internal or non-public validation datasets, which restricts the generalizability of the results globally[5,7,11]. Furthermore, such datasets may harbor biases that are not evident until deployment, potentially resulting in unfair and discriminatory AI healthcare outcomes, and limiting efficient integration in existing colposcopy systems. A limited number of preclinical AI research studies have addressed these concerns, such as external validation using open-source datasets to assess the generalizability of a model in new environments[12].

The WHO/IARC has recently released an open-source standardized colposcopy image bank to address the critical needs of cervical cancer screening, by providing a standardized resource for training healthcare professionals and enhancing global cervical cancer screening and early

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Table 1 | CIN2+ Diagnosis Performance of AI System and Colposcopists

	Sensitivity% (95% CI)	Specificity% (95% CI)	PPV% (95% CI)	NPV% (95% CI)
AI system	84.2 (74.4–90.7)	55.9 (46.6–64.7)	56.6 (47.4–65.4)	83.8 (73.8–90.5)
All readers	84.2 (81.3–87.0)	58.0 (52.1–63.9)	59.5 (55.9–63.2)	83.9 (81.6–86.2)
Low-experienced readers	78.6 (74.7–83.2)	58.3 (54.3–62.0)	57.0 (50.6–63.0)	79.5 (74.4–84.5)
High-experienced readers	88.5 (85.8–91.1)	56.0 (51.6–60.3)	58.5 (52.6–64.1)	87.3 (83.8–91.1)

PPV Positive Predict Value, *NPV* Negative Predict Value, *CI* Confidence Interval.

diagnosis capabilities[13]. This dataset contains diverse cervical images captured during colposcopy examinations from various clinical settings, accompanied by expert diagnoses. It facilitates the development of innovative tools such as AI-based systems to improve cervical cancer detection and serves as a valuable resource for external validation of AI algorithms. The colposcopy image bank provides a spectrum from precancer to cancer cases for evaluating the generalizability and robustness of these models across various populations and clinical scenarios.

In this study, we expanded the retrospective dataset and integrated an independent WHO open-source dataset for external testing of an AI-guided colposcopy system for detecting CIN2+ and guiding biopsy efficiency[13]. The Food and Drug Administration (FDA) recommends a multiple-reader multiple-case (MRMC) study design as a validation method for assessing the benefit of clinical AI solutions[14], which assesses the performance of a set of colposcopists interpreting images under different modalities, with and without AI. We aimed to compare its diagnostic performance with those of colposcopists and study whether the performance of colposcopists improves when they are provided with AI assistance.

Results

AI system vs. colposcopist diagnostic performance

A total of 187 cases were included in this study, 111 cases were scored as benign and CIN1, and 76 cases were scored as CIN2+ histology. Reader-level diagnostic performance metrics are summarized in Table 1. The AI system achieved a sensitivity of 84.2% (95% CI 74.4–90.7%) and a specificity of 55.9% (95% CI 46.6–64.7%) for CIN2+ detection. Across the 30 colposcopists, the overall reader performance showed a similar sensitivity of 84.2% (95% CI 81.3–87.0%) and a specificity of 58.0% (95% CI 52.1–63.9%). However, performance differed by experience level. The sensitivity of low-experienced colposcopists was significantly lower than AI system (78.6%, 95%CI 74.7–83.2% vs. 84.2%, 95%CI 74.4–90.7%). 21 high-experienced colposcopists showed a much higher sensitivity (88.5%, 95%CI 85.8–91.1% vs. 78.6%, 95%CI 74.7–83.2%) and a lower specificity (56.0%, 95%CI 51.6–60.3% vs. 58.3%, 95%CI 54.3–62.0%), compared to 9 low-experienced colposcopists. Figure 1 presents confusion matrix heatmaps for the AI system (case-level) and pooled colposcopist assessments (read-level), illustrating CIN2+ detection error patterns stratified by experience level. We also performed a sensitivity analysis restricted to the Lugol-free subset ($n = 42$; 28 CIN2+ and 14 < CIN2). In this modality-matched subset, the overall performance pattern remained consistent with the primary analysis. The AI system achieved sensitivity comparable to high-experienced colposcopists, suggesting that its diagnostic accuracy is driven by acetic acid imaging and does not depend on Lugol's iodine (Supplementary Table S1).

AI-assisted vs. unassisted colposcopist performance and variability

By the end of two reading phases, 15 colposcopists completed the interpretation of the colposcopic images, with and without AI system in a total of 5,610 reading records. AI assistance numerically improved discrimination for CIN2+ across all readers, with AUC increasing from 72.8% (95% CI, 68.4–77.1%) without AI to 76.5% (95% CI, 72.9–80.1%) with AI; however, the overall difference did not reach statistical significance (ΔAUC = 3.7%, 95% CI, −0.1–7.5%). Low-experienced colposcopists ($n = 11$) had AUC changes in AI-assisted and unassisted scenarios of 72.4% (95% CI, 67.5–77.3%) to 76.3% (95% CI, 72.3–80.3%), corresponding to a ΔAUC of 3.9% (95% CI, 0.1% to 7.8%; $p = 0.043$), which was significantly improved in diagnosis with AI assistance. For high-experienced colposcopists ($n = 4$), the AI-assisted AUC was also numerically higher than the unassisted AUC, increasing from 73.8% (95% CI, 67.9–79.8%) to 76.9% (95% CI, 71.8–81.9%) (ΔAUC = 3.0%; 95% CI, −7.5–13.5%). The overall performance for colposcopists with and without AI assistance was visualized in AUC-ROC curve (Fig. 2).

The detection of sensitivity in CIN2+ by all colposcopists was significantly increased with AI assistance (90.6%, 95%CI 88.7–92.1% vs. 84.8%, 95%CI 82.7–86.9%). In the low-experienced colposcopist group, the diagnostic sensitivity rose 6.5% (88.7%, 95%CI 86.4–90.7% vs. 82.2%, 95%CI 79.4–84.6%) by assisting with the AI system. The specificity was also significantly increased (63.9%, 95% 61.2–66.5% vs. 62.6%, 95%CI 59.8–65.2%), but not in high-experienced colposcopists. The diagnosis performance of colposcopists with and without AI assistance was shown in Table 2. When AI assisted in the diagnosis of all colposcopists, the accuracy was increased to 73.6% (95% CI 71.9–75.2%) from 70.4% (95%CI 68.7–72.0%). Kappa value for high-experienced colposcopists was higher than for low-experienced colposcopists, and showed an improvement across the two scenarios in both experienced-level colposcopists. The overall performance of AI-assisted colposcopist diagnosis on CIN2+ was improved when compared with colposcopist interpretation.

Biopsy concordance and reader confidence

Only 76 CIN2+ cases were asked by colposcopists to indicate no more than four biopsy sites with and without AI-guided biopsies recommendation. The average number of biopsies by the case was reduced from 2.48 to 2.02 with AI assisted. For the 76 cases evaluated, the PPB was increased in the scenario of AI-assisted diagnosis by low-experienced colposcopists alone (Median [IQR]: 0.86 [0.48–1.00] vs. 0.80 [0.37–1.00]; $p = 0.0305$). The high-experienced colposcopists show a not significant increase of PPB, since the higher inter variability. All colposcopists showed an increase in confidence in diagnosis with AI assistance (69.4 vs. 77.8), was shown in Fig. 3. The confidence boost was particularly pronounced in low-experienced colposcopists, with an improvement of 10 points (from 67.4 to 77.4), while high-experienced colposcopists showed a 5-point increase (from 74.9 to 79.1). We conducted a subgroup analysis on confidence regarding CIN2+ cases, and the results showed that low-experienced colposcopists still had a significant confidence improvement (from 66.8 to 77.5), while high-experienced colposcopists' confidence increased by about 2 points (from 75.1 to 76.9).

Discussion

In our publicly available dataset based multireader multicase external validation study across Chinese various experience-level colposcopist, the effect of AI-assisted diagnosis performance of colposcopic examination was significantly improved. This study resulted the generalization capability of AI-guided colposcopic system, CAIADS, and statistical clinical evaluation of AI-assisted diagnostic capacity. CAIADS demonstrated promising accuracy in enhancing high-grade lesion detection rates and reducing the risk of missed diagnoses, which was similar to few previous studies[5,9,15]. Regarding its independent diagnostic performance, the AI system demonstrated higher sensitivity in detecting CIN2+ cases but exhibited lower specificity when evaluated on the WHO/IARC dataset. In view of this external validation, the

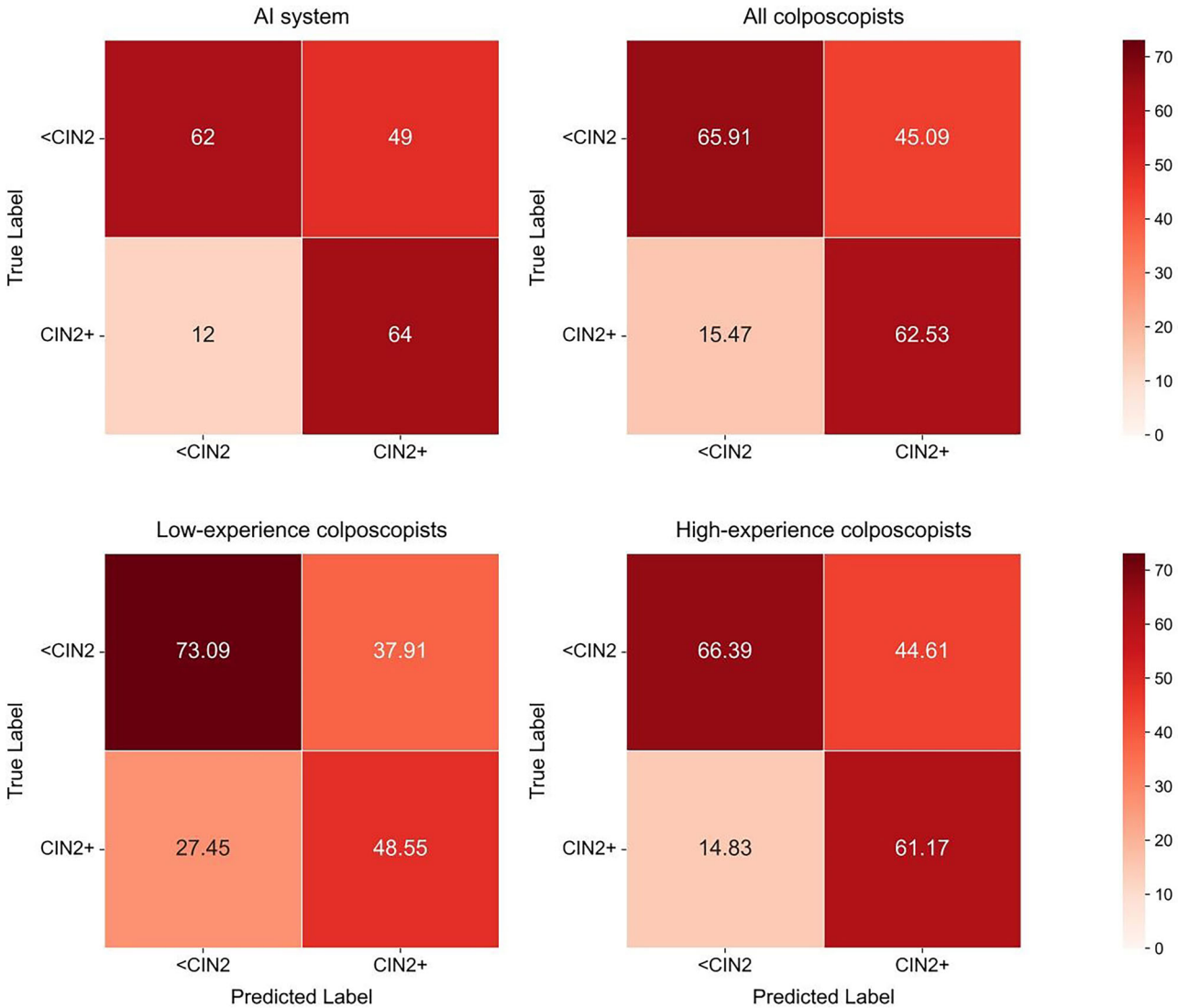


Fig. 1 | Confusion-matrix heatmaps for CIN2+ detection. The AI system is evaluated as a single reader at the case level (n = 187). Colposcopist heatmaps summarize the pooled distribution of reader-case assessments (read level) for all readers and for low- and high-experience subgroups.

AI system is comparable in its ability to detect CIN2+ to that of colposcopists with approximately five years of experience. It identified 113 cases as positive with 84.2% sensitivity, but only 64 were confirmed as true positives, with a significant 43.4% of normal or low-grade lesions misclassified as CIN2 + . When trained, the model was optimized to prioritize sensitivity to avoid missing high-grading cases. The standalone AI system showed slightly lower specificity than colposcopists (55.9% vs 58.0%), suggesting a higher false-positive rate despite its high sensitivity for CIN2+ detection. This tradeoff may lead to unnecessary biopsies and reflects the current limitation of AI in more refined lesion-grade discrimination. However, the high negative predictive value (83.8%) indicates good reliability for ruling out CIN2+ disease, which may support low-risk stratification. Notably, AI assistance improved specificity in both low-experienced and high-experienced readers, with a significant improvement in low-experienced readers and a similar increasing trend in high-experienced readers. These findings suggest that, although the standalone AI model may overcall some abnormalities, its integration into clinical assessment can still improve the specificity of colposcopic decision-making by helping readers standardize interpretation and avoid biopsy of less relevant lesions. Therefore, there seems to be a persisting internal bias, possibly resulting from the model's reliance on certain features, such as brightness, contrast, or color variations, as discriminating factors. Designed external validation studies serve to evaluate the generalization capability of the CAIADS and offer evident insights to guide further optimization by identifying its limitations. The diagnostic results of a number of colposcopists using an AI-guided colposcopy system in this study demonstrated the potential of this system to prevent missed diagnoses in different clinical settings.

There are several limitations to this study. First, the publicly available image data sets were utilized for diagnosis with the lack of relevant primary screening information, where colposcopists were only able to simulate the colposcopic examination. In real clinical practices, women who have abnormal hr-HPV test or cytology results are typically referred for colposcopic examination. Many colposcopic findings and diagnostic criteria need to take into account the results of hr-HPV testing or cytology. However, the advantage of using public data is its reproducibility for other researchers. Future studies should aim to integrate CAIADS into colposcopes from various vendors and evaluate its performance in real-time within a multicenter clinical setting by a prospective study design. Secondly, the MRMC study design offers improvement in the statistical power and objectivity of clinical studies by mitigating biases caused by variations in reader experience in which diagnostic AI equipment is used. It reduces sample size requirements by incorporating multiple readers, which accounts for inter-reader variability and enhances the statistical efficiency of the trial. However, reading order bias is the potential limitation, where readers may be

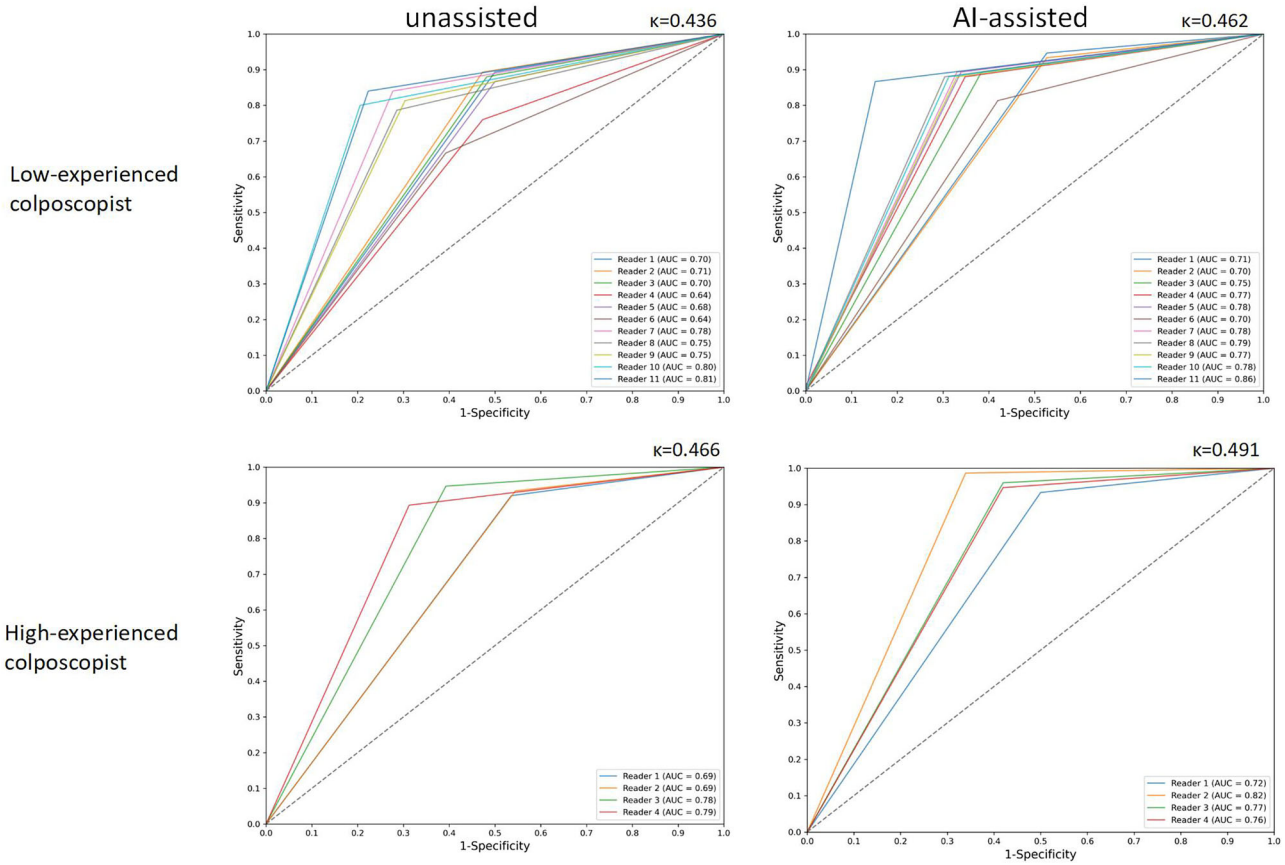


Fig. 2 | ROC analysis of CIN2+ detection with and without AI assistance. Receiver operating characteristic curves compare reader performance under unassisted and AI-assisted conditions, and Fleiss'κ is reported to quantify interreader agreement.

influenced by information from previous cases and diagnostic modes, despite efforts to randomize the reading order. Additionally, confirmation bias may occur in MRMC studies where binary test results are preferentially selected for gold-standard validation, potentially skewing the assessment of diagnostic performance, especially in retrospective studies. Additionally, our study focused exclusively on evaluating the diagnostic performance of AI-assisted colposcopists for detecting CIN2+ cases. Considering the limited diagnostic skills of colposcopists in low- and middle-income countries (LMICs), AI support is the emphasis on enhancing the detection of high-grade lesions within a short period and improving cervical cancer screening rates. However, the classification of CIN1/CIN2 also carries critical implications for clinical management and decision-making. Due to CIN2 possibly turning negative, accurate specificity indicates a potential overdiagnosis and over-biopsy reduction. Thus, the capability of AI to differentiate between CIN1, CIN2, and CIN3 requires rigorous evaluation and scientific validation. The robustness of this study is grounded in its methodological design. By leveraging an independent public dataset alongside a MRMC framework, we ensured high transparency while rigorously accounting for inter-reader variability. First, an independent publicly available dataset was used, which improves the reproducibility and transparency of the study and allows benchmarking under the same conditions. In addition, the comparison between standalone and AI-assisted reading under the same framework allowed a direct assessment of the incremental value of AI assistance across readers with different experience levels.

In the cervical cancer screening pathway, colposcopy is a critical triage step for identifying high-grade lesions and guiding targeted biopsies[16]. We selected CIN2+ as the primary endpoint because it represents a clinically actionable threshold for intervention, aligning with the intended use of our AI-assisted system[17]. Importantly, this system serves as an adjunctive tool to improve the consistency of risk stratification and standardize biopsy site selection, particularly where operator experience is limited. The core clinical value of AI application lies in maximizing the detection of critical precancerous lesions while minimizing both missed high-grade disease and unnecessary invasive sampling.

While CIN2+ is a reasonable primary endpoint for assessing the performance of our AI-assisted colposcopy system, we acknowledge the importance of distinguishing between CIN2 and CIN3+ in clinical practice. Identifying CIN2 lesions is crucial because, while they may progress, they also have a high rate of regression, especially in younger women (less than 30 years of age). This characteristic makes the management of CIN2 lesions more complex, as the decision to treat or monitor is influenced by various factors, including the patient's age, HPV status, and reproductive plans[18–20]. By contrast, CIN3+ lesions typically warrant immediate intervention due to their higher progression risk to invasive cancer. Thus, clinical stratification between CIN2 and CIN3+ is essential for optimizing management strategies. This specific diagnostic challenge is exactly why CIN2+ is a relevant, real-world endpoint for evaluating an AI colposcopy-support tool.

Still, claims regarding clinical utility must remain grounded in the current dataset. Our findings support the use of AI for biopsy guidance, but do not yet confirm improved long-term management outcomes. Future work should focus on system refinement and prospective evaluation of its ability to discriminate finer lesion categories under clinically relevant workflows.

A major advantage of AI colposcopy is its potential to narrow accuracy disparities in healthcare and training facilities with highly limited. The main finding of our study was that colposcopists can benefit significantly from the diagnostic results generated by the CAIADS for identical cases, leading to an improvement in diagnostic performance. This effect was especially prominent among various experience-level colposcopists. As hypothesized, the

Table 2 | The performance of AI-assisted and unassisted colposcopist on CIN2+ diagnosis

	Without AI	With AI	*P*-value[a]
Accuracy, % (95% CI[b])			
All readers	70.4 (68.7–72.0)	73.6 (71.9–75.2)	–
Low-experienced readers	70.4 (68.4–72.4)	73.8 (71.8–75.7)	–
High-experienced readers	70.2 (66.8–73.4)	73.1 (69.8–76.2)	–
Sensitivity, % (95% CI[b])			
All readers	84.8 (82.7–86.9)	90.6 (88.7–92.1)	–
Low-experienced readers	82.2 (79.4–84.6)	88.7 (86.4–90.7)	–
High-experienced readers	92.3 (88.8–94.8)	95.6 (92.7–97.5)	–
Specificity, % (95% CI[b])			
All readers	60.7 (58.3–63.0)	62.3 (60.0–64.6)	–
Low-experienced readers	62.6 (59.8–65.2)	63.9 (61.2–66.5)	–
High-experienced readers	55.4 (50.7–59.9)	58.0 (53.4–62.5)	–
PPV, % (95% CI[b])			
All readers	59.1 (56.7–61.5)	61.6 (59.3–64.0)	–
Low-experienced readers	59.5 (56.6–62.3)	62.2 (59.4–64.9)	–
High-experienced readers	58.1 (53.6–62.4)	60.4 (56.0–64.7)	–
NPV, % (95% CI[b])			
All readers	85.7 (83.5–87.6)	90.8 (89.0–92.3)	–
Low-experienced readers	84.0(81.5–86.2)	89.4 (87.2–91.3)	–
High-experienced readers	91.5 (87.6–94.3)	95.2 (92.0–97.2)	–
AUC, % (95% CI[b])			
All readers	72.8 (68.4–77.1)	76.5 (72.9–80.1)	0.057
Low-experienced readers	72.4 (67.5–77.3)	76.3 (72.3–80.3)	**0.043**
High-experienced readers	73.8 (67.9–79.8)	76.9 (71.8–81.9)	0.427

PPV Positive Predict Value, *NPV* Negative Predict Value, *AI* Artificial Intelligence.
[a]*p*-values are calculated only for the primary endpoint (AUC) via the iMRMC framework to handle correlated reads.
[b]Secondary binary metrics are evaluated descriptively using case-level bootstrap 95% CIs. Low-experienced readers: 3–5 years of experience in colposcopy; High-experienced readers: > 5 years of experience in colposcopy.

diagnostic AUC of all colposcopists improved by 3% with AI assistance. For relatively colposcopists with 3-5 years of experience, the improvement of sensitivity was statistically significant. In previous studies[7,9,21], AI-assisted clinicians' diagnoses have consistently demonstrated higher sensitivity than when they worked alone, suggesting a potential reduction in missdiagnosis. Similarly, in our study, this outcome was observed in the different experienced colposcopist groups. The results revealed that AI assistance resulted in increased sensitivity for low-experienced colposcopists that was superior to AI alone, yet still lower than that of high-experienced colposcopists. However, our findings showed that high-experienced colposcopists had lower specificity and that AI-assisted diagnosis did not result in significant improvements. It suggests that AI support may prompt them to apply stricter criteria when identifying <CIN2 cases. This could be due to their efforts to minimize missed diagnoses may have resulted in an overlooked risk of overdiagnosis. We initially hypothesized that CAIADS assistance would enhance the ability of junior colposcopists to detect HSIL cases[22], but we found it also assisted the experienced colposcopists in better diagnosis performance with stronger confidence.

One of the primary objectives of colposcopy is to guide biopsies for pathological confirmation, aiming to select an adequate number of biopsy sites to ensure diagnostic accuracy while minimizing suboptimal biopsies[23]. CAIADS was trained on a large dataset of annotated biopsy images to enable real-time inference of biopsy site locations[22]. This study is the first to evaluate colposcopists' biopsy performance with and without AI assistance, demonstrating a significant reduction in the average number of biopsies per case. The evaluation of biopsy efficacy focused on the accuracy of site selection, defined as the proportion of biopsy sites aligned with standard reference directions. This metric primarily reflects the precision of sampling direction but does not directly address whether critical biopsy sites within the standard reference sites were missed. Due to the lack of site-specific pathological outcomes in the WHO/IARC dataset, biopsy efficacy was assessed at the case level, using the proportion of accurate biopsy sites among all sites indicated per case. The primary metric assessed colposcopist accuracy, defined as the proportion of biopsy sites aligning with standard reference directions by WHO/IARC. In the AI-assisted scenario, the precision of biopsies performed by junior colposcopists improved significantly. The average number of biopsies per case decreased significantly, which indicated a more efficient biopsy for low-experienced colposcopists. While this approach highlights the precision of sampling direction, it does not directly address the potential omission of critical biopsy sites within the standard reference areas. Therefore, a prospective study design to evaluate the biopsy accuracy with site-level histologic results is necessary for assessing AI-guided colposcopy system guiding biopsy efficiency.

Publicly available datasets offer insights into the performance, generalizability, and potential applications of AI models to ensure the transparency of experiments and reproducibility of results, facilitating comparative analysis across different studies and advancing innovative technological progress in the field. Due to the limitation of the public colposcopic image dataset, AI colposcopic models were developed on privately recruited colposcopic images. AI models typically require large numbers of high-quality data to achieve comparable diagnosis performance. However, private datasets often face significant challenges regarding the standardization and authenticity of data collection, which directly impacts the training effectiveness of AI models. Although many AI models are developed on extensive private images[5,8,24], the lack of external validation studies makes it difficult to determine whether these models are effective in the clinics. Furthermore, several studies have demonstrated that AI systems show considerable potential in detecting between HSIL and LSIL[6,25–27]; however, the appropriate treatment decision is needed to determine further clarification of the lesion grades. Private datasets may lack sufficient diversity to effectively train AI models to accurately classify cervical intraepithelial neoplasia grades. Early pilot studies by Yuan et al. [5] demonstrated the diagnostic accuracy of the AI model for cervical lesions and validated by a collected set of images; however, further validation in large-scale prospective clinical study is still limited. In addition to improving the credibility and robustness of AI colposcopy models, publicly available datasets are evident for enhancing the model's adaptability and reliability.

In conclusion, this proof-of-concept diagnostic accuracy study using an independent WHO/IARC dataset demonstrated that AI assistance was associated with improved detection of high-grade cervical intraepithelial neoplasia and potential reduction of unnecessary biopsies, with the greatest benefit observed among less experienced colposcopists. However, these preliminary findings require rigorous validation in large-scale, multicenter prospective trials before clinical utility and generalizability can be established.

Methods

Materials and study dataset

In this study, an analysis of 200 cases was collected based on a publicly accessible dataset consisting of 913 colposcopy images provided by WHO/IARC[28]. Data and metadata were accessed on August 28, 2024. Datasets were

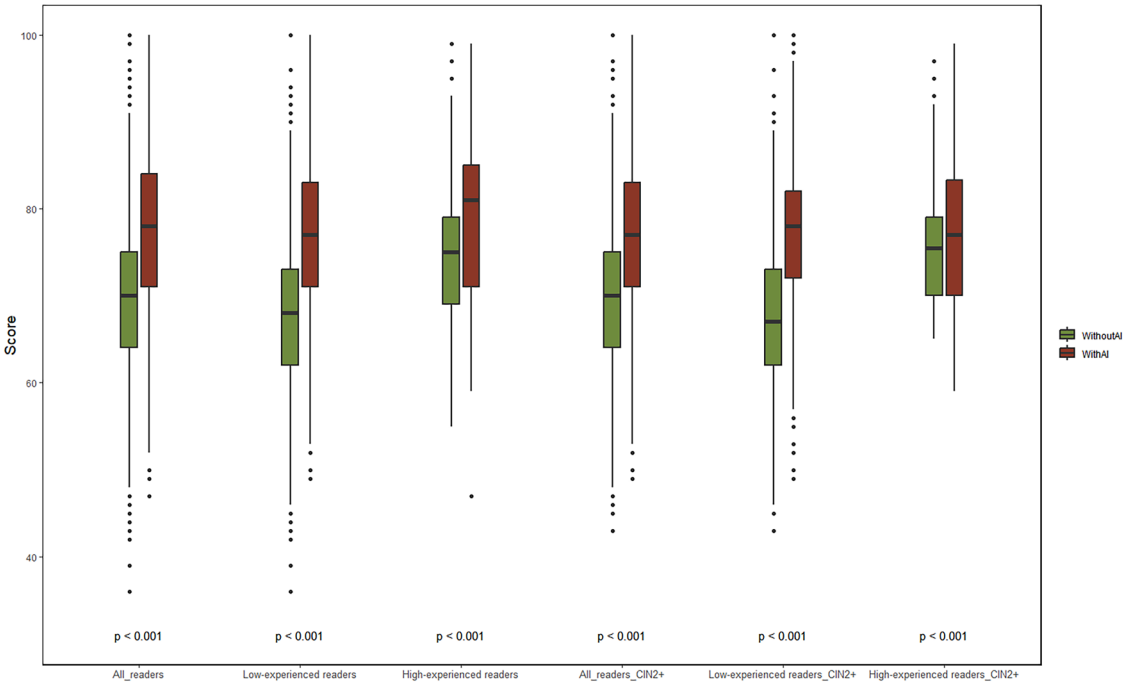


Fig. 3 | Boxplots of colposcopists' confidence scores for all cases and CIN2+ cases with and without AI assistance.

selected by their rigorous quality control process, which involved systematic image collection by collaborating colposcopists using standardized protocols, followed by collation and validation. In the AI inference process, colposcopic images with Lugol's solution applied to the cervix were not included, but images with acetic acid and Lugol's solution applied to the cervix were included for colposcopist interpretation. Additionally, 13 cases of inadequate colposcopy due to poor focus and insufficient post-acetic acid applied images were excluded from the study. The remaining 855 colposcopy images from the 187 cases were grouped into 76 cases of CIN2+ and 111 cases of <CIN2 of the histopathology findings. The International Federation for Cervical Pathology and Colposcopy (IFCPC) terminology was used to determine colposcopic impressions[29]. The CIN classification system was used for histologic evaluation. Histological results were categorized as normal/benign or CIN1 were classified as <CIN2, while those categorized as CIN2, CIN3, or cancer were classified as CIN2 + . In cases involving lesions varying in severity across different sites, the diagnosis was determined considering the most severe lesions. This study was approved by the Institutional Review Board of the Chinese Academy of Medical Sciences and Peking Union Medical College (No. CAMS&PUMC-IEC-2024-028), and was conducted in accordance with the Declaration of Helsinki. The study used de-identified publicly available data from the WHO-IARC cervical image bank. Written informed consent was obtained from all colposcopists participating in the reader study before participation.

Reader study

Fifty colposcopists with different experience levels were recruited from 12 regions in China to evaluate their CIN2+ diagnosis ability with and without AI diagnostic system. All readers are recruited from local obstetrics gynecology hospitals, with 3–5 years and more than 5 years of colposcopic examination experience. Informed consent was obtained from all colposcopists at the time of enrollment, and they were asked to indicate their intention to participate in either the human-AI comparison study or the AI-assistance MRMC study. According to the European Federation for Colposcopy (EFC), performing ≥50 colposcopies annually is considered the minimum requirement for a qualified colposcopist; therefore, only colposcopists meeting this criterion were eligible[30].

Figure 4 shown the entire study design. In human-AI comparison study, a total of 30 colposcopists independently assessed all colposcopy images of cases with each primary screening result (HPV test), with no access to clinical information, colposcopy reports, and histopathological results during the assessment. All readers were required to report on colposcopy adequacy, squamocolumnar junction (SCJ) visibility, transformation zone category, colposcopic diagnosis, and biopsy selection. In AI-assistance MRMC study, a group of 15 colposcopists completely participated in a two-phase reading process was conducted on the online assessment platform, with training on platform operations provided before the study for each phase. Based on common practices in diagnostic imaging research[31,32], a 2-week washout period was incorporated between the two phases of the study to minimize the potential for recall bias. In the first phase, all colposcopists reviewed images of each case and recorded the colposcopy findings, primary diagnosis, and biopsy sites blinded to ground truth and without AI. After two weeks of washout, each colposcopist reinterpreted all cases in a randomized order with the assistance of AI to prevent any systematic bias due to the case presentation order or previous readings. For each case, all participants were asked to rate their confidence in diagnoses in two phases with a maximum of 100 considered as "very confident". The MRMC study design was described in reader assignments in Fig. 5. Five recruited colposcopists who did not meet two trail requirements or poor quality performance were excluded. Overall, 45 colposcopists participated across two trials, of which 20 low-experienced level colposcopists with 3-5 years of experience and 25 high-experienced level colposcopists with more than 5 years of experience. In the human-AI comparison study ($n = 30$), there were 9 low-experienced and 21 high-experienced colposcopists. In the AI-assistance MRMC study ($n = 15$), there were 11 low-experienced and 4 high-experienced colposcopists.

AI-guided colposcopy system

The AI system evaluated in this study was developed previously and has been described in detail[22]. The Colposcopic Artificial Intelligence Auxiliary Diagnostic System (CAIADS)[22] was initially developed and validated on high-resolution digital colposcopic images and non-image information and consisted of several deep-learning modules for cervix detection, lesion diagnosis grading, and biopsy sites detection. The detection deep-learning module was based on CNN-ResNet-50 as the backbone for extracting cervix, whereas the lesion grade classification module was designed on Graph Convolutional Network (GCN)-based architecture for fusing

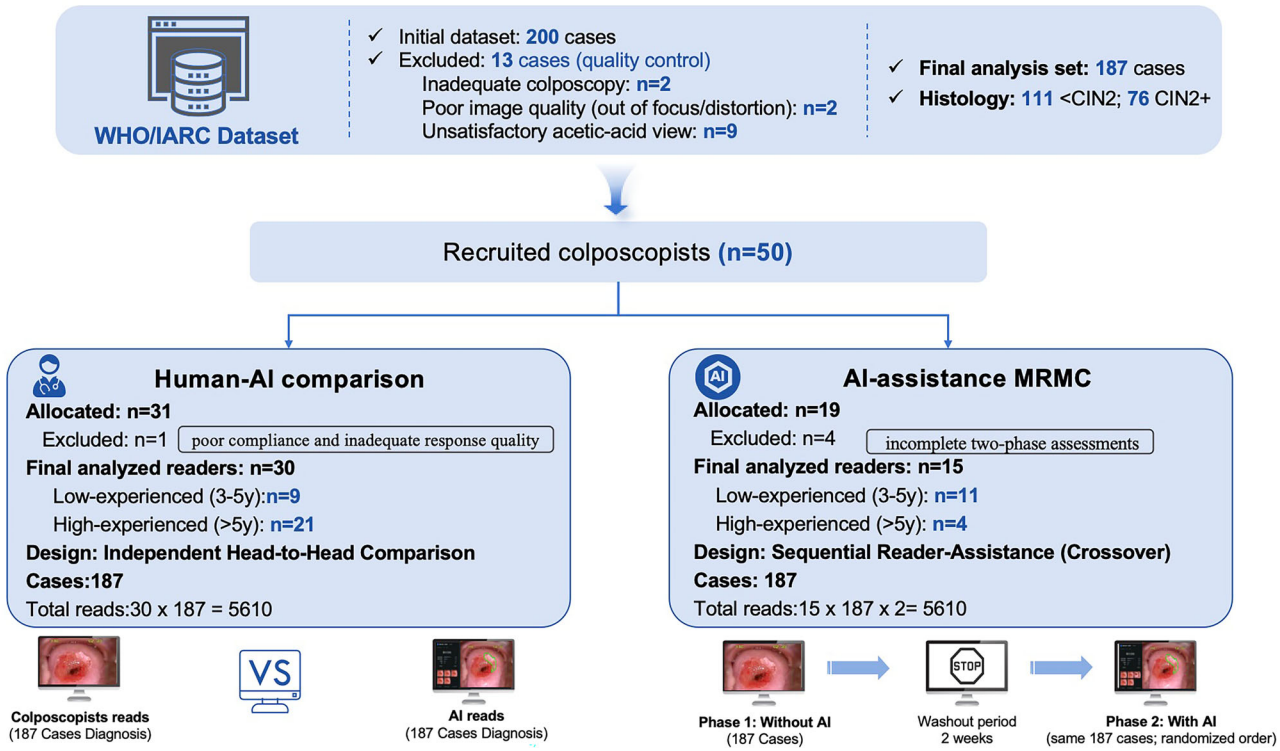
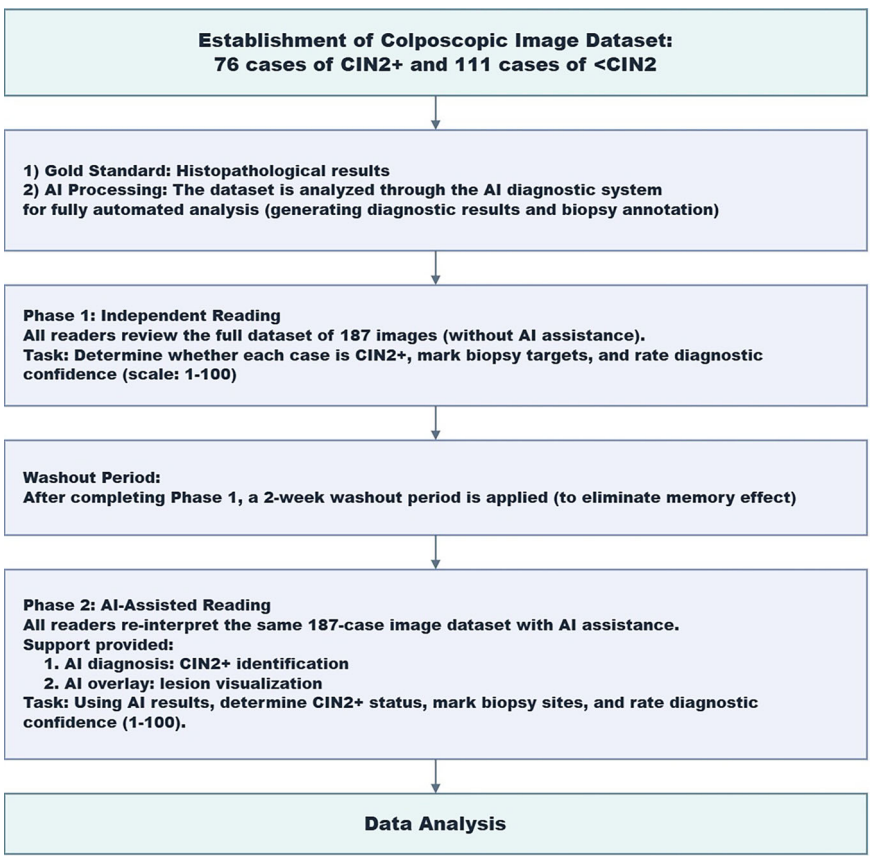


Fig. 4 | Study design. Each step in the study modules depicts case distribution, colposcopist diagnosis, reading tasks with and without AI implementation, and the main framework of data analysis.

Fig. 5 | MRMC reader study flowchart.



features. The annotated lesion location of colposcopic images was trained for a segmentation module based on U-Net5 and YOLOv3 to address the biopsy guidance. The AI system was trained by 13,000 multicenter patients with approximately 60,000 digital colposcopic images and non-image information. It achieved 82.2% accuracy for lesion grading with histological results as the gold standard on a test set of 3000 patients, and its standalone diagnosis sensitivity was higher than colposcopist. It also presented the accurate biopsy site prediction with a 75.8% distribution of intersection on ground truth biopsy sites. For the current independent external validation, CAIADS was executed using a strictly fixed inference configuration and the same prespecified decision rules established in the original development study. The system processes all eligible colposcopic images available for a given case (typically ~5 images per case) to generate a single case-level binary diagnosis (CIN2+ vs. < CIN2) alongside biopsy guidance. Crucially, no model tuning, threshold adjustment, or recalibration was performed on the WHO/IARC dataset; all inference parameters were locked a priori. During the AI-assisted reading phase, the evaluation platform displayed the CAIADS case-level diagnostic output and biopsy guidance (comprising lesion visualizations and suggested biopsy areas with red bounding box).

Statistical analysis

Two reader studies were conducted and analyzed separately; the diagnosis performance was assessed for both. The sample size of interpretation cases and number of readers were determined based on variability of readers and patient diversity[33]. Performance evaluation was conducted using histopathological results as the gold standard. The primary endpoint was histologic HSIL (CIN2+ vs. <CIN2). For each reader and the AI system, we calculated sensitivity, specificity, accuracy, positive predictive value (PPV), and negative predictive value (NPV) based on the CIN2+ classification. 95% confidence intervals (CIs) for these binary metrics were obtained using nonparametric bootstrap resampling at the case level[34].

In the AI-assistance MRMC study, the prespecified primary comparison was the difference in area under the receiver operating characteristic curve (ΔAUC) for CIN2+ detection between AI-assisted and unassisted reads across all colposcopists. To properly account for the correlated nature of the data across readers and cases, AUCs, ΔAUC, and their corresponding 95% CIs and p-values were estimated using the iMRMC R package. For secondary binary metrics, including sensitivity, specificity, PPV and NPV, differences were evaluated descriptively, with 95% CIs derived from case-level nonparametric bootstrapping to appropriately capture variance. Inter-reader agreement for lesion grading was assessed using Fleiss κ. The sensitivity analysis restricted to the Lugol-free cases subset was additionally performed, aligning with the AI system's exclusive reliance on this modality for inference.

For each case, readers additionally provided a diagnostic confidence score on a 1–100 scale (100 = very confident). Changes in confidence scores between unassisted and assisted phases were evaluated by paired t-test. Biopsy concordance was quantified by the proportion of biopsies matching the reference biopsy clock-position(s) provided in the WHO/IARC dataset metadata. Given the absence of site-specific histopathological outcomes in the dataset, biopsy performance was assessed at the case level using the proportion of precise biopsy sites (PPB). For each case, PPB was defined as: PPB = (number of reader-indicated biopsy clock-position(s) concordant with the reference clock-position(s) provided in the dataset metadata)/(total number of biopsy sites indicated by the reader for that case). PPB values were then summarized across cases for comparisons between reading phases. Paired comparisons of biopsy concordance between reading phases were assessed using the Wilcoxon signed-rank test. Analyses were performed in R (v4.4.2) and Python (pandas, scikit-learn). A two-sided p value < 0.05 was considered statistically significant.

Data availability

The data used in this study are de-identified and publicly available from the International Agency for Research on Cancer cervical image bank at the following website: https://screening.iarc.fr/cervicalimagebank.php

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Author contributions

P.X. and Y.Q. conceptualised and designed the study. T.W. and Y.W. monitored the study implementation, data collection, data entry into the software, and help two different groups of participants with specific software usage problem; Y.H., X.W., and J.W. were responsible for model development and inference; T.W. and Y.W. performed data cleaning and statistical analysis, and wrote the initial draft of the manuscript; X.C. and Y.Q. provided supervision throughout the study; P.X. and Y.Q. conducted critical revisions of the manuscript for key content. The final decision to submit the manuscript for publication was made by all authors.

Competing interests

The authors declare no competing interests.

Additional information

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